Programs on Decision Making and Knowledge Representation

a) Introduction to PROLOG

Aim:

To study the basics of PROLOG (facts, rules and queries) and write simple programs using SWI-Prolog.

Procedure:

1. Open Notepad (or the SWI-Prolog editor).
2. Write the facts and rules and save the file with **.pl** extension.
3. In SWI-Prolog, load the program using **File -> Consult** and select the file.
4. Type the query at the **?-** prompt, ending it with a full stop (.). Use **;** to get further answers.
5. Observe and record the output.

Prolog Code:

Exercise 1: Family Relationships

Problem: Define family relations and answer grandparent queries.

```
% Facts
parent(john, mary).
parent(mary, susan).
parent(john, mike).

% Rules
grandparent(X, Y) :- parent(X, Z), parent(Z, Y).

% Query
% ?- grandparent(john, susan).
```

Output:

true

Exercise 2: Animal Classification

Problem: Identify if an animal is a mammal based on features.

```
% Facts
has_hair(tiger).
gives_milk(tiger).

% Rule
mammal(X) :- has_hair(X), gives_milk(X).

% Query
% ?- mammal(tiger).
```

Output:

```
true.
```

Exercise 3: Simple Arithmetic Knowledge Base

Problem: Implement simple addition using PROLOG.

```
% Rule
add(X, Y, Z) :- Z is X + Y.

% Query
% ?- add(5, 7, Z).
```

Output:

```
Z = 12.
```

Result:

Thus the basic PROLOG programs using facts, rules and queries were written and executed successfully.